

# Ethics, Bias, Privacy & Predictive Modeling Report

## Anti-Money Laundering Data Analytics Case Study at Pig E. Bank

This report combines three key components of the Pig E. Bank analytics project:

- Linear regression analysis
- Ethics and bias evaluation
- Privacy and security risk assessment

The goal is to not only interpret analytical results but also evaluate the responsible use of data in a banking environment.

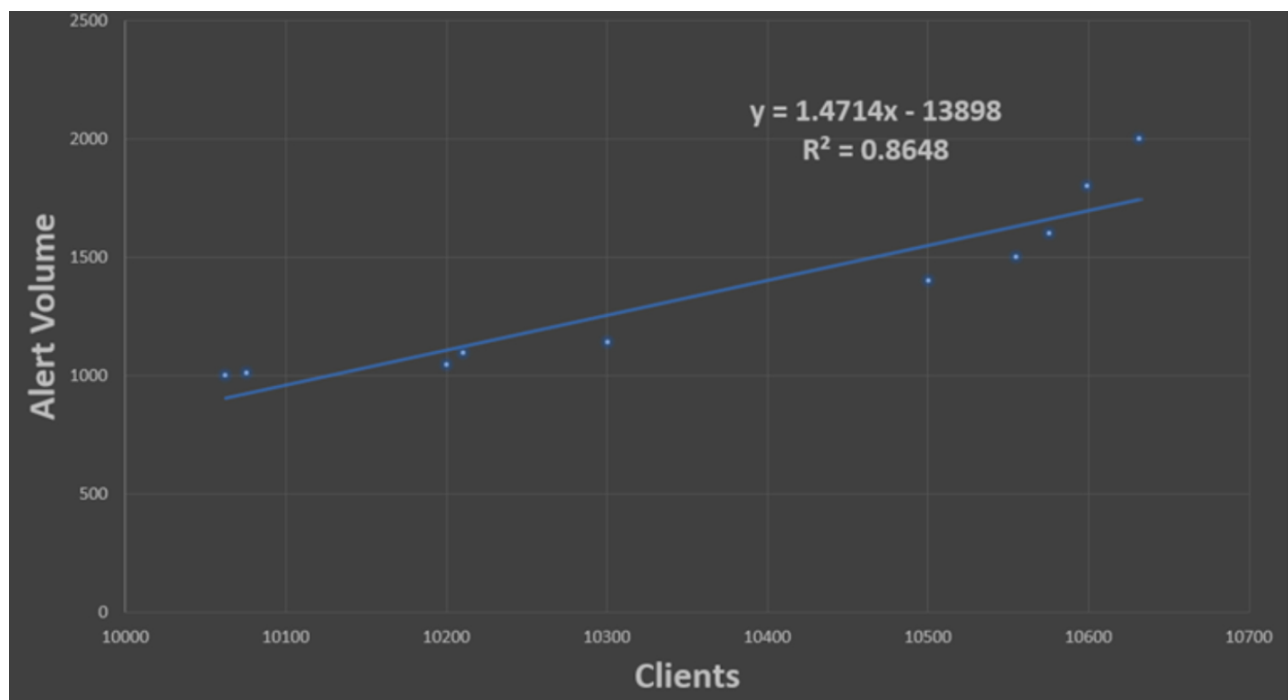
## 1. Linear Regression Analysis

### 1.1 Objective

The objective of the regression analysis was to examine the relationship between customer base size and fraud alert volume.

### 1.2 Method

A linear regression model was applied using historical operational data to identify whether increases in customer numbers were associated with changes in fraud detection activity.



### 1.3 Key Results and Interpretation

- A strong positive relationship was observed between customer base size and fraud alert volume. For every one additional client the bank gains, the model predicts an increase of approximately 1.47 alerts. This indicates operational dependency rather than randomness in fraud detection events.
- The model demonstrated strong explanatory power with an  **$R^2$  of approximately 0.86**. This means approximately 86% of the variation in alert volume can be explained by the number of clients, suggesting the model is suitable for forecasting workload and operational planning.
- Residual analysis indicated generally consistent model fit, since the data points are visually clustered very closely to the regression line. While there is some slight "noise", there are no extreme outliers that would suggest the model is being skewed. While more clients lead to more alerts, a sudden spike in alerts could also be caused by a new type of cyberattack, regardless of client count. We would always need to monitor those residuals for sudden shifts.

Assessment: The model has high fitness. It is a robust tool for Pig E. Bank to use for capacity planning. For instance, if the bank expects to hit 11,000 clients next month, they can plug that number into the equation to predict alert volume and ensure they have enough investigators on staff to handle the load.

## 2. Ethics & Bias Evaluation

Because the model relies on human labeling and historical transaction data, it is highly sensitive to bias in data collection, sampling, and measurement.

### 2.1 Data Collection Bias

- Certain customer groups may be underrepresented in the dataset, this can lead to skewed insights and reduced model fairness.
- Suppose we only included alerts that were confirmed as fraud and ignored the "false positives." The result would be that the model would only learn what "obvious" fraud looks like. It wouldn't help the bank predict the total alerts, which was the original goal.

### 2.2 Sampling Bias

- Non-random sampling may distort the representation of customer behavior patterns. This affects the generalizability of churn and fraud predictions.
- If we only collected data from "high-net-worth" clients because their records were easier to access or more detailed, our model would be biased. The result would be that the regression might predict alert volumes perfectly for rich clients but fail miserably for student accounts or small businesses, as their transaction patterns (and fraud risks) are different.

### 2.3 Labeling & Measurement Bias

- Inconsistent investigator judgments can lead to unreliable labels and differences in interpretation introduce noise into training data. This reduces model accuracy.
- If the software used to count "alerts" had a glitch where it double-counted alerts during peak hours or failed to record them if the system was lagging. As a result, the alert volume would be artificially high or low, leading to a regression line that doesn't reflect reality.

### 2.4 Impact of Bias

- Bias leads to inaccurate fraud detection thresholds
- Certain groups may be over-flagged
- False positives increase workload for investigators
- True suspicious activity may be missed
- Reduces trust and fairness in automated alert systems

Bias directly impacts financial crime detection systems by distorting who is flagged as suspicious and how alerts are generated.

### **3. Privacy & Security Risk Assessment**

The primary risks relate to internal misuse, external exposure, and insufficient anonymization.

#### **3.1 Data Access Risks**

- Poorly managed access permissions can increase risk of misuse
- Internal users may unintentionally or intentionally expose sensitive data

#### **3.2 Data Leakage Risks**

- Unauthorized data capture (e.g., screenshots, exports) poses security threats
- Weak control around data sharing can increase exposure risk

#### **3.3 Outsourcing Risks**

- External analytics partners introduce regulatory complexity
- Cross-border data transfer may conflict with privacy regulations

#### **3.4 Data Anonymization Limitations**

- Poor anonymization techniques may still allow re-identification

## **4. Risk Mitigation Recommendations**

### **4.1 Data Governance & Bias Control**

- Implement training and standardized labeling guidelines for investigators
- Conduct regular bias audits on datasets and models
- Use randomized and representative sampling across all customer groups
- Include both positive and negative cases (including false positives)
- Cross-check internal data with external benchmarks

### **4.2 Privacy Protection Measures**

- Enforce strict role-based access control
- Prevent unauthorized data extraction and storage
- Strengthen anonymization methods before external sharing

### **4.3 Security Enhancements**

- Monitor and log all data access activity
- Restrict sensitive dataset permissions to essential personnel only

### **4.4 Compliance & Operational Improvements**

- Ensure alignment with data protection regulations (e.g., GDPR principles)
- Improve documentation of data workflows and model decisions
- Perform regular internal audits of analytics processes

## **5. Conclusion**

This report highlights that while predictive modeling can provide valuable operational insights, its reliability depends heavily on data quality, fairness, and governance. The linear regression model effectively supports operational forecasting, but ethical and privacy considerations remain critical to ensure responsible use of banking data. Addressing bias, strengthening privacy controls, and improving security governance are essential to maintaining trust and compliance in financial analytics systems.